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Module 3 Introduction

Lesson 1: Linear Regression of an Indicator Matrix

Lesson 2: Linear Discriminant Analysis (LDA)

Reading: Linear Discriminant Analysis (LDA) Readings
45 min

Video: Linear Discriminant Analysis (LDA)
9 min

Practice Assignment: Linear Discriminant Analysis (LDA) Quiz
Submitted

Lab: Coding Example
2h

Lab: Coding Exercise
1h

Lesson 3: Logistic Regression

Module 3 Summative Assessment

Module 3 Summary

Your grade: 100%

Next item

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Linear Discriminant Analysis (LDA) Quiz

Review Learning Objectives

Instructions

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

1. What is the primary goal of Linear Discriminant Analysis (LDA)? **1 / 1 point**

- To predict the value of a continuous target variable.
- To reduce the dimensionality of data while preserving as much variance as possible.
- To cluster observations without prior knowledge of group membership.
- To classify observations into distinct categories by maximizing the variance between categories.

Correct
Correct!

Submitted Nov 16, 12:13 AM IST

Attempts Unlimited

Try again

2. Which assumption does LDA make about the distribution of features in each class? **1 / 1 point**

Your grade
100%

View submission See feedback

- Features follow a Poisson distribution.
- Features are independent and identically distributed.
- Features in each class follow a Gaussian distribution with class-specific mean and a common covariance matrix.
- Features follow a uniform distribution.

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Correct
Correct!

3. What does LDA use to separate different classes? **1 / 1 point**

- The mean values of features.
- The difference in variance within classes.
- A linear combination of features that maximizes the ratio of between-class variance to within-class variance.
- The median values of features.

Correct
Correct!